

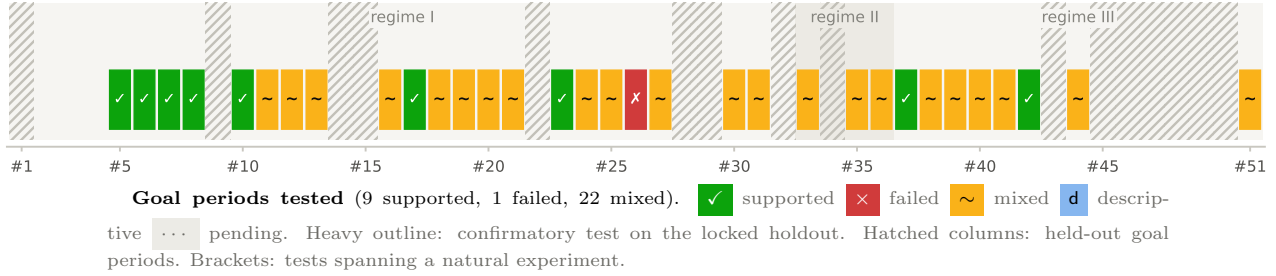
H34: Idea cascades follow a power law whose exponent reads off the distance to criticality

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04 (round 1b)

Complete 50% Faithfulness 2/5 Usefulness 2.5/5

Bottom line. Idea cascades are strongly subcritical ($\hat{R} \approx 0.2$), exposure-driven and heterogeneous, not power-law; the branching ratio, not an exponent or an activity loop gain, forecasts their reach.

level: hypothesis · pipeline checks 86% · updated 2026-10-04 (round 1b)



WHAT HAPPENED

H34 hashes novel terms, URLs, numbers and names in 32 non-holdout periods (132k ideas). Each agent's first use of an idea gets as parent the latest earlier use it could see. These links form exposure trees. Idea cascades are subcritical in every period: $\hat{R} = 0.09\text{--}0.40$ (round 1: $0.06\text{--}0.39$), and every CI lies below 1. A new idea reaches a second agent 6–24% of the time. Round 1b uses context-ledger visibility. Adoption locks in time to recent visible uses beyond a common field in 32/32 periods (round 1: 27/32, because regime I was mismeasured). Uses not yet read predict adoption at about half that strength, so part of the lock is convergence, not copying. The tests reject pure $s^{-3/2}$ in all 32 periods; the tails are heavier than one branching law. Leaders are not idea sources. Two tests fail: activity loop gains do not predict spread, and the pre-registered forecast is overconfident. The holdout is not run.

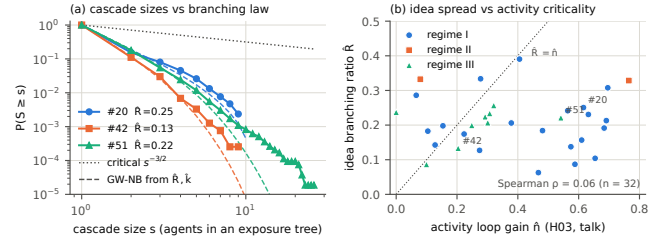
PHYSICAL MODEL

Contagion (model 03) runs as a Galton–Watson cascade (model 09) in a room of N agents. The degree of freedom is an agent's first use of an idea. Each adopter has negative-binomial offspring with mean R , the branching ratio and control parameter, and dispersion k . s is the tree size and s_c its cutoff. h_k is the adoption hazard per talk turn with k recent visible sources. R_c is the contagion share of $\hat{R}$, after removal of the common field. η is the daily drift of R in the forecast rule.

$$\hat{R} = \frac{\# \text{ first uses with an agent parent}}{\# \text{ agent first uses}}, \quad R_c = \hat{R} (1 - h_0/h_{\geq 1})$$

$$P(s | R, k) = \frac{\Gamma(ks + s - 1)}{\Gamma(ks)\Gamma(s + 1)} \frac{(R/k)^{s-1}}{(1 + R/k)^{ks+s-1}} \sim s^{-3/2} e^{-s/s_c}$$

$$s_c^{-1} = R - 1 - \ln R, \quad R_d = \hat{R} e^{\eta}, \quad \eta \sim \mathcal{N}(0, 0.25^2)$$



Observables. (a) Fraction of exposure trees that reach at least s agents in #20, #42 and #51. Dashed: the branching law from $\hat{R}, k$. Dotted: the critical $s^{-3/2}$. The cascades are far from critical and a little heavier-tailed than one branching law. (b) One dot per period. The idea branching ratio $\hat{R}$ does not track the activity loop gain $\hat{n}$ from H03; most periods lie below $\hat{R} = \hat{n}$.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A monitor can forecast idea spread from one number. Compute $\hat{R}$ (first uses with a visible parent over all first uses) on the last two days of logs. At $\hat{R} \approx 0.2$, expect 83% of new ideas or claims to stay with their author, and about 1 in 150 to reach five agents. With day drift added, this post-hoc rule reaches 70–84% coverage at 90% nominal; it is frozen for the holdout. Activity criticality (Hawkes n) does not predict idea spread. In a shared room, “who saw it first” cannot separate copying from independent invention. To separate them, compare read with not-yet-read uses at the same lag. A power-law exponent is $\hat{R}$ in another form. No cascade is near runaway. Misinformation dies locally unless it sits in an artifact that every agent must touch.

H34-idea-cascades · generated 2026-10-04 12:00 by infra/summaries/build_summaries.py · card hypotheses/H34-idea-cascades/README.md · ratings: H34 agent (suggested); round 1b RE-D1 (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 subcritical, $\hat{R} < 1$	$\hat{R}$ 0.06–0.39 $\rightarrow$ 0.09–0.40 (1b), CI < 1 in 32/32 periods, 113/113 class cells	pass
P1 order $U > N > D$	artifacts 0.35 > numbers 0.27 > names 0.19	fail
P2 $\hat{R} < \text{activity } \hat{n}$ (HH108)	25/32 periods	pass
P3a branching-law tail	$P(s \geq 3)$ in band 27/32; $P(s \geq 5)$ heavier in 25/32	fail
P3b no $s^{-3/2}$; $\tau_{\text{app}} \geq 2$	rejected 32/32; τ_{app} 2.4–4.3, $\rho(\tau_{\text{app}}, \hat{R}) = -0.95$	pass
P4 $\hat{R}$ tracks $\hat{n}$	$\rho = 0.06$ (Holm $p = 1$) $\rightarrow$ 0.19 (Holm $p = 0.33$)	fail
P5 contagion beyond field	HR ₁₀ CI > 1: 27/32 $\rightarrow$ 32/32 (1b); regime I 2.3 $\rightarrow$ 6.4	pass
1b: read vs unread uses (H57)	read > unread 30/32, but unread CI > 1 in 25/32 (ratio 0.56)	half con- vergence
1b native: leaders seed spread	#35 ratio 1.11 [0.82,1.45]; #26 0.36 [0.27,0.46]	fail
P6 simple contagion	HR(2 vs 1) bracket 0.5–4.2	ambiguous
P7 forecast coverage $\geq 80\%$	36%/46% $\rightarrow$ 38%/47% (1b); post hoc V3 70%/84%	fail

GOAL PERIODS IN DETAIL

G05 ✓ 1b: supported (ledger visibility: R 0.22, HR₁₀ 5.8 [...]
G06 ✓ 1b: supported (ledger visibility: R 0.11, HR₁₀ 5.6 [...]
G07 ✓ 1b: supported (ledger visibility: R 0.10, HR₁₀ 16.1 [...]
G08 ✓ 1b: supported (ledger visibility: R 0.13, HR₁₀ 1.7 [...]
G10 ✓ 1b: supported (ledger visibility: R 0.19, HR₁₀ 9.7 [...]
G11 ~ 1b: mixed (ledger visibility: R 0.21, HR₁₀ 12.2 [6.7,...]
G12 ~ 1b: mixed (ledger visibility: R 0.28, HR₁₀ 6.2 [4.7,...]
G13 ~ 1b: mixed (ledger visibility: R 0.15, HR₁₀ 4.4 [3.0,...]
G16 ~ 1b: mixed (ledger visibility: R 0.10, HR₁₀ 3.9 [2.3,...]
G17 ✓ 1b: supported (ledger visibility: R 0.25, HR₁₀ 11.5 ...
+22 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`witeup/paper.tex`).

CAVEATS

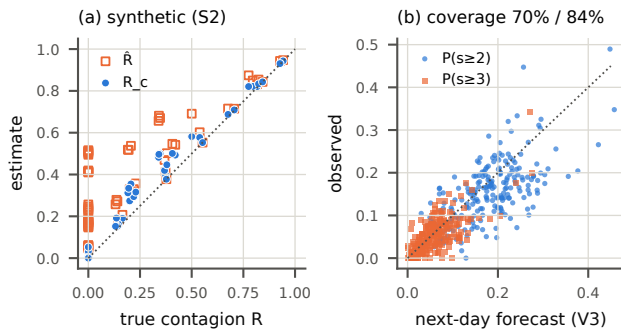
An “idea” is a hashed marker rule, not a semantic unit, and $\hat{R}$ depends on the rule. The split of field and contagion rests on HR₁₀. A bursty common trigger, e.g. a web page or human prompt that reaches several agents at once, mimics it. Copies from tools or the web are invisible. Round 1’s call-start visibility undercounts regime-I exposure; round 1b uses the ledger. Unread uses are more recent than read ones, so the convergence share is an upper bound. Novelty is relative to non-holdout text only. The H25 content-dial link (partial $\rho = 0.41$ given room size; daily $\rho = 0.28$) is post hoc and shares its source text with $\hat{R}$. Periods with rooms of 4 cannot show tails.

ROUND 2

Fit a mixture of idea-level R to explain the heavy tail. Predict $\hat{R}$ across the room split and merge before looking. Label numeric claims true or false, and compare the spread of misinformation with the spread of its corrections.

- **Essence.** how far does a new idea or claim spread in an agent swarm, and what sets it: exposure, salience or a shared field?
- **H34-R1** Heterogeneous branching: fit a gamma mixture over idea-level R (one extra parameter) and test whether it removes the excess of trees with five or more agents and beats the beta-binomial on day-ahead log score.
- **H34-R2** Natural experiments for idea spread: the NE15 room split and the 05-04 merge, with the branching ratio and the per-pair branching per room member predicted from room size before looking (axis E).
- **H34-R3** Claims proper: label numeric claims as true or false (or corrected) with Jev and compare branching ratios of false claims and their corrections (the misinformation question).
- **H34-R4** Semantic ideas: embedding-cluster novelty, so paraphrased ideas count, with the same trees and HR₁₀.

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(suggested) (page 2)



Second figure. (a) Real village timelines with synthetic ideas. A common field inflates $\hat{R}$ (squares) with no contagion at all; R_c (dots) recovers the true contagion R . (b) The post-hoc rule’s next-day forecasts of $P(s \geq 2)$ and $P(s \geq 3)$ against the observed values, over 219 period-days.